Feature Selection of Diabetic Retinopathy Disease Using Particle Swarm Optimization and Neural Network

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***Abstract*- Currently, there are many medical experts who face difficulty in conducting early detection for diabetic retinopathy. This occurs because it is difficult to recognize the early symptoms of this disease. In order for this disease to be detected early, an accurate classification method is required. Data mining concept is one alternative in conducting classification. This study was conducting by applying particle swarm optimization (PSO) method to select the best Diabetic Retinopathy feature based on diabetic retinopathy dataset. Then, the selected feature is further classified using classification method of neural network. The study result show that there is an increase in result by applying neural network based particle swarm optimization (PSO) of 76.11%. This study also show that there is an increase in classification result by using feature selection method of 4.35% from previous result of 71.76% by only applying neural network method.**

Keywords – ***Diabetic Retinopathy, Classification, particle swarm optimization (PSO), Neural Network***

I. INTRODUCTION

Diabetes is one of diseases that result in death. Data of World Health Organization (WHO) shows that 1.6 millions of deaths in 2015 were caused by this diabetes disease [1] and Indonesia is the country with the largest diabetics in Asia Pacific region [2]. Most of diabetics in Indonesia are type 2 (Non-Insulin Dependent Diabetes Mellitus) caused by unhealthy life style patterns and usually affects humans within 30-60 year age range. One of diabetes later known as the cause of blindness is diabetic retinopathy. Diabetic retinopathy is caused by insulin levels that continue to increase in the blood, and this disease is one type of disease that is progressive that early treatment is required to prevent blindness [3]. The early stage of diabetic retinopathy will not directly cause blindness to patient, if the cancer turns into advanced stadium, patient will experience several symptoms such as blurry vision and periodic eye ache. If these symptoms are not treated immediately, patients will experience blindness [4].

Given this fact, it is necessary to conduct early detection of diabetic retinopathy in order to minimize the risk of blindness. In order to perform early detection of diabetic retinopathy, studies have been conducted by many experts. In 2014, Balint Antal and Andras Hajdu conducted a study on retinal image data obtained from screening result of patients that later is known as dataset of messidor. The method used in this study is *ensemble learning* implemented using *machine learning classifier*. The result of the study shows the levels of sensitivity of 90%, *specificity* of 91%, accuracy of 90% and AUC (area under the curve) of 0.989 [5]. Next study was conducted in 2016 by Carmen Valverde, Maria Garcia, Roberto Hornero, and Marial Lopez Galvez. In this study, the problem discussed is the large variability on data basis of that inhibits direct comparison of different study and researches. This encourages Valverde, Garcia, Hornero and Galvez to summarize detection and classification algorithm based on automatic retinal image analysis. The result of this study is the comparison between dataset of diabetes retinopathy conducted to several algorithm methods including *k*-NN, Fuzzy C-Means, Neural Network, Support Vector Machine and Multylaer Perceptor[6]. Both of previous researches put forward the image analysis in each research, while this study will generate classification percentage of dataset in numerical form so that accuracy level obtained can be known.

In this study, to maximize numerical classification result of retinopathy, classification optimization would be conducted by applying feature selection using Particle Swarm Optimization (PSO) method. Similar researches that use feature selection have been done previously by experts. For example, a study conducted by Herliana in 2016, where in her study, optimization was conducted by applying Correlation based Features Selection (CFS) method towards single cell of Pap Smear dataset [7]. Feature selection was also applied in a study by Hidayatulloh, Herliana and Arifin in 2016. In this study, dataset of Pap Smear's single cell was used as study object with accuracy level reaches 95.7% for 2 classes and shows improvement from the study conducted by Herliana in 2016 [8]. In addition, feature selection by utilizing PSO method itself has been done by several experts, such as in a study by Arifin in 2017 who conducted feature selection towards Pap Smear single cell dataset [9]. In the study, PSO method was applied to generate data training and selected

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testing, selected data training will be tested by applying classification data mining method. Besides Arifin, PSO method was also applied by Zhang, Gong and Cheng in 2017. In this study, several dataset were utilized as result comparator from PSO method combined with Hybrid Mutation method. The result of this study shows that the combination of 2 methods could provide advantage above 0.8 on Metric Hyper Volume (HV) of 5 datasets tested [10].

Besides utilizing feature selection, in this study, classification is also conducted using Neural Network (NN) method. Previously, classification method by utilizing NN method has been applied by many experts for different objects than this study. One of the studies utilizing NN method as its classification method is the study by Praveena and Singh in 2015. In this study, classification method was conducted to classify image data taken from satellite, then the images were segmented using Artificial Bee Colony (ABC) method [11]. Besides image data, classification method is also used in coronary artery disease datasets performed by Radhimeenakshi in 2016. The study compares Neural Network and Support Vector Machine methods toward coronary artery disease datasets of Cleveland and Statlog (binomial class and polynomial class). The study results shows that Neural Network shows better score in Metric Specify than Support Vector Machine method [12].

II. DIABETIC RETINOPATHY

Diabetes retinopathy is a disease that attacks the eye of diabetic patients. This disease can cause blindness if not treated immediately. According to Lubis, diabetes retinopathy is a progressive microangiopathy characterized by damage and blockage of capillaries [13]. Given the fact that retina of the eye in its activity requires blood flow to work well, it is understandable why this disease attacks diabetes patients. Diabetes retinopathy itself is classified into two types, Non Proliferative Diabetic Retinopathy and Proliferative Diabetic Retinopathy. Detection of the existence of diabetes retinopathy is currently conducted with a physical examination by trained ophthalmologist. This method is quite effective but requires many involved experts. Since the number of diabetic patients keeps increasing, so does the number of DR patients, therefore physicians require more time to perform detection for different patients [14].

To make it easier for physicians in detecting this disease early, researchers have succeeded in forming a dataset known as diabetic retinopathy Debrecen dataset [15] that is also used as reference in this study. It is known that dataset used in this study is 1151 data. Below are Table 2 and 2 that describe the datasets used.

TABLE 1
Attributes in Diabetes Retinopathy datasets

Name	***Information***
Attribute 0	*The binary result of quality assessment*
Attribute 1	*The binary result of pre-screening*
Attribute 2-7	*The results of MA detection*
Attribute 8-15	*Exudates are represented by a set of points rather than the number of pixels constructing the lesions*
Attribute 16	*The euclidean distance of the center of the macula and the center of the optic disc to provide important information regarding the patient's condition*
Attribute 17	*The diameter of the optic disc*
Attribute 18	*he binary result of the AM/FM-based classification*
Attribute 19	*Class label. 1 = contains signs of DR (Accumulative label for the Messidor classes 1, 2, 3), 0 = no signs of DR*
Class	*1 = contains signs of DR (Accumulative label for the Messidor classes 1, 2, 3), 0 = no signs of DR.*

TABLE 2
Number of Datasets based on Class

NO	Type of Classification	Number of *Record Dataset*
1	*Contains signs*	611
2	*No signs*	540
Total		1151

III. METHODS

A. Particle Swarm Optimization

Particle swarm optimization (PSO) is an optimization algorithm introduced by Dr. James Kennedy and Dr. Russell C Eberhart in 1995. This algorithm was inspired by life pattern of bird and fish population. The advantage of optimization method of *Particle Swarm Optimization* is that it has a simple concept, easy to implement, and is efficient in calculation than other math algorithm and heuristic optimization techniques [9].

The image of PSO itself is implemented by Kennedy and Eberhart by description of search formula of velocity and position of a particle:

$$v_i(t) = v_i(t-1) + \varphi C_1(p_i - x_i(t-1)) + \varphi C_2(g - x_i(t-1))$$

$$x_i(t) = x_i(t-1) + v_i(t)$$

Detail:
X_i = candidate for particle completion
P_i = experience of best particle position
g = information of best particle environment

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$X_i(t-1)$ = current particle position
φ = random vector with score range of 0.1
C1 = coefficient of *cognitive learning*
C2 = coefficient of *social learning*

B. Neural Network

Artificial neural network method or *Neural Network* was introduced by Waren McCulloch and Walter Pitts in 1943. It is called *Neural Network* because the shape of this method working system resembles neuron cell system of human brain. The working concept of *Neural Network* or better known as *Artificial Neural Network* (ANN) is similar with the performance of neural network of human brain [11]. More fully, the structure of ANN is described as follow:

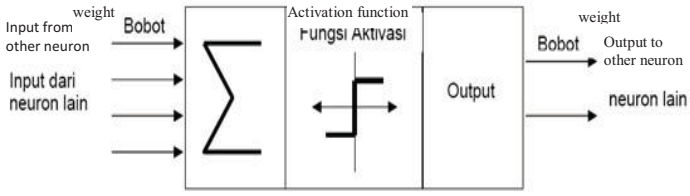


Fig 1. Structure of ANN

Based on Figure 1, it is known that the characteristics of ANN can be seen from the connection of one neuron to other. The process starts when input is obtained from other neuron that will be followed by processing by *summing function* where in it occurs the summing process of input value that will be compared with certain *threshold*. If the summing result is larger than *threshold,* activation process will be cancelled, on the other hand is the summing result is smaller that *threshold* value, neuron will be activated. Once the neuron is active, next step is output process to transmit activated weight values of the neuron to be continued to other neuron. This series of process will continue on the next neuron input.

C. Proposed Method

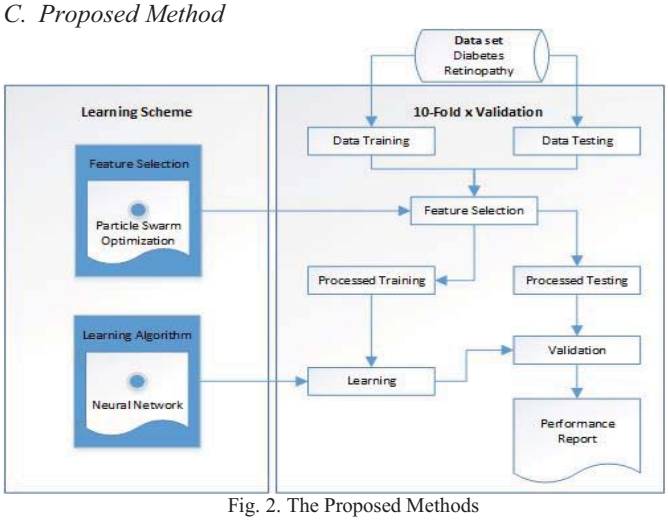


Fig. 2. The Proposed Methods

Based on the description of the proposed methods in Figure 2, research flow is explained by the following steps:

a. Starting from dividing of diabetic retinopathy dataset into 10 parts using 10 *fold cross validation*, all parts from dataset will become training data and testing data. Once divided, every feature or attribute is selected using Particle swarm optimization method.
b. After best feature or attribute is obtained, the training data is classified using neural network method.
c. Meanwhile, testing data whose features have been selected is validated based on training data and method used in this validation is also neural network.
d. After training data and testing data is validated using neural network, the last step is the presentation of data classification result.

IV. RESULT AND DISCUSSION

This stage explains experiment results using *Neural Network* and *Neural Network* + Particle Swarm Optimization algorithm. Accuracy value for *Neural Network* method is 71.76% and *Neural Network + Particle Swarm Optimization* method is 76.11%

TABLE 3
Algorithm Experiment Result of *Neural Network* on Diabetes Retinopathy Dataset

Accuracy: 71.76%			
	True No Sign	*True Contains Signs*	*Class Precision*
Pred. *No Sign*	382	167	69.58%
Pred. *Contains Signs*	158	444	73.75%
Class Recall	70.74%	72.67%	

TABLE 4
Algorithm experiment result of *Neural Network + Particle Swarm Optimization* on Diabetes Retinopathy datasets

Accuracy: 76.11%			
	True No Sign	*True Contains Signs*	*Class Precision*
Pred. *No Sign*	453	188	70.67%
Pred. *Contains Signs*	87	423	82.94%
Class Recall	83.89%	69.23%	

Once overall accuracy value of each method is known, next step is to compare *accuracy, precision* and *recall* values. The following table describes the comparison between *accuracy, precision* and *recall* values.

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TABLE 5
Comparison of Accuracy, Precision and Recall Values

	Neural Network	*Neural Network + Particle Swarm Optimization*
Accuracy	71.76%	76.11%
Precision	75.07%	83.81%
Recall	72.67%	69.22%

Next, calculation result is visualized by ROC curve. Comparison result can be seen in the following table.

TABLE 6
Experiment Result Comparison

Method	Accuracy	AUC
Neural Network	71.76%	0.803
Neural Network + Particle Swarm Optimization	76.11%	0.827

Table 6 shows *accuracy* and AUC of each method, it is seen from the highest *accuracy Neural Network + Particle Swarm Optimization* value, as well as AUC value, and it can be concluded that the methods tested is classified as good because they have AUC values between 0.80-0.90.

V. CONCLUSIONS

Based on previous experiment in feature selection using *Neural Network-based Particle Swarm Optimization* to classify Diabetic Retinopathy, the conclusions include:

1. From research result, the highest score of AUC is obtained, that is 0.827 and accuracy score is 76.11% by using Neural Network model with feature selection of Particle Swarm Optimization, while for experimental result on Neural Network model, AUC score obtained is 0.803 and accuracy level is 71.76%.
2. Result of the whole experiments shows that there is an increase in Neural Network model using feature selection of Particle Swarm Optimization compared to Neural Network model as many as 4.35%.

Future researches can be conducted by using Image Processing technique with retina image as the object to detect diabetes, and accuracy can be improved by applying other algorithm such as Particle Swarm Optimization, C.45, and Naïve Bayes.

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2. Data used in this study was obtained from dataset of diabetic retinopathy Debrecen

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